

MALTA LUQA, Malta

WMO: 165970

Lat: 35.857N

Lon: 14.478E

Elev: 91

StdP: 100.23

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.9	7.9	-0.1	3.8	9.9	1.8	4.3	10.4	14.1	12.2	12.4	12.0	3.5	290	0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	34.2	21.4	32.8	21.8	31.2	22.0	25.3	28.8	24.7	28.2	24.2	27.7	3.6	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.3	19.4	26.6	23.8	18.9	26.4	23.1	18.0	26.2	77.9	28.5	75.4	28.3	73.2	27.7	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.4	10.1	9.0	4.7	37.8	1.0	2.2	4.0	39.3	3.4	40.6	2.8	41.8	2.1	43.4		
(5)			2.5	26.6	1.2	1.0	1.7	27.3	1.0	27.8	0.4	28.4	-0.5	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	19.3	12.7	12.4	13.9	16.2	19.8	24.0	26.8	27.2	24.6	21.4	17.6	14.3	(6)
(7)		DBStd	5.65	1.97	2.11	2.07	1.96	2.16	2.61	1.91	1.78	1.95	2.07	2.19	2.08	(7)
(8)		HDD10.0	8	3	4	1	0	0	0	0	0	0	0	0	1	(8)
(9)		HDD18.3	722	174	166	139	68	9	0	0	0	0	2	38	126	(9)
(10)		CDD10.0	3398	87	71	121	186	304	422	521	534	437	353	229	134	(10)
(11)		CDD18.3	1071	0	0	1	4	54	172	262	276	187	97	17	1	(11)
(12)		CDH23.3	8291	0	0	4	12	216	1279	2542	2756	1198	273	10	0	(12)
(13)	CDH26.7	2795	0	0	1	1	35	425	1014	1052	246	21	0	0	(13)	
(14)	Wind	WSAvg	4.1	4.5	4.7	4.7	4.8	4.4	3.7	3.4	3.2	3.6	3.7	4.1	4.6	(14)
(15)	Precipitation	PrecAvg	517	80	57	38	20	7	3	0	5	36	88	81	100	(15)
(16)		PrecMax	826	196	171	100	68	43	36	1	129	208	254	411	234	(16)
(17)		PrecMin	242	11	9	0	0	0	0	0	0	0	8	0	18	(17)
(18)		PrecStd	168	53	40	24	19	10	8	0	24	46	60	92	64	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.6	19.2	22.1	25.0	30.0	35.8	37.0	36.8	32.2	28.9	24.6	20.8	(19)
(20)			MCWB	15.2	14.8	14.9	16.4	18.1	20.1	21.6	21.5	22.7	22.3	20.1	16.8	(20)
(21)		2%	DB	17.3	17.8	19.7	22.4	27.2	32.8	34.2	34.1	30.1	26.9	23.1	19.1	(21)
(22)			MCWB	14.5	14.3	14.6	15.6	17.9	20.6	21.3	22.1	22.7	21.7	19.3	15.7	(22)
(23)		5%	DB	16.8	16.8	18.2	21.0	25.8	30.9	32.8	32.6	29.0	25.9	22.1	18.2	(23)
(24)			MCWB	14.2	13.9	14.2	15.4	17.5	20.3	21.5	22.5	22.7	21.2	18.7	15.2	(24)
(25)		10%	DB	16.0	15.9	17.1	19.9	24.1	29.0	31.2	31.1	28.0	24.9	21.1	17.6	(25)
(26)			MCWB	13.4	13.2	13.8	15.1	17.3	20.0	21.8	22.8	22.4	20.8	17.8	14.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.7	15.7	16.2	17.7	20.7	24.0	25.6	26.4	25.7	24.3	21.5	17.5	(27)
(28)			MADB	17.6	17.8	19.6	21.8	24.8	28.9	30.3	29.6	28.2	26.1	23.3	19.4	(28)
(29)		2%	WB	15.1	14.9	15.5	16.9	19.7	23.0	24.7	25.6	24.9	23.3	20.2	16.5	(29)
(30)			MADB	16.9	16.8	18.2	20.5	23.9	27.3	28.8	29.0	27.6	25.2	22.4	18.4	(30)
(31)		5%	WB	14.5	14.2	14.9	16.2	19.0	22.2	24.1	25.0	24.3	22.5	19.2	15.7	(31)
(32)			MADB	16.3	16.2	17.3	19.3	22.9	26.7	28.3	28.7	26.8	24.6	21.5	17.7	(32)
(33)		10%	WB	13.8	13.5	14.3	15.7	18.3	21.5	23.5	24.4	23.6	21.6	18.4	15.0	(33)
(34)			MADB	15.6	15.5	16.5	18.5	22.2	26.1	27.8	28.2	26.3	24.0	20.8	17.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.3	5.7	6.2	6.6	7.7	8.4	8.7	7.9	6.6	6.1	5.5	5.1	(35)
(36)			MCDBR	5.6	6.2	7.2	8.4	9.8	10.5	10.8	9.8	7.6	6.8	6.2	5.4	(36)
(37)			MCWBR	3.9	4.2	4.1	4.0	4.5	4.9	5.6	5.2	3.8	3.6	4.0	3.8	(37)
(38)		5% WB	MCDBR	5.2	5.8	6.5	7.0	8.2	9.1	9.0	8.2	6.8	6.1	5.8	5.1	(38)
(39)			MCWBR	4.1	4.3	4.1	3.7	4.2	4.6	5.0	4.4	4.1	4.0	4.4	3.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.354	0.375	0.427	0.466	0.467	0.443	0.451	0.453	0.451	0.424	0.388	0.360	(40)	
(41)		taud	2.445	2.369	2.200	2.109	2.150	2.262	2.252	2.255	2.263	2.308	2.370	2.434	(41)	
(42)		Ebn at Noon	835	858	838	823	826	844	833	825	805	793	785	801	(42)	
(43)		Edh at Noon	88	106	136	157	152	136	137	134	126	111	94	84	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.56	3.46	4.78	6.02	7.09	7.73	7.80	6.90	5.38	3.89	2.77	2.28	(44)	
(45)		RadStd	0.16	0.25	0.27	0.29	0.33	0.30	0.13	0.26	0.20	0.23	0.19	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.18	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	-0.27	-0.30	N/A	N/A	N/A	N/A

Nomenclature: See separate page